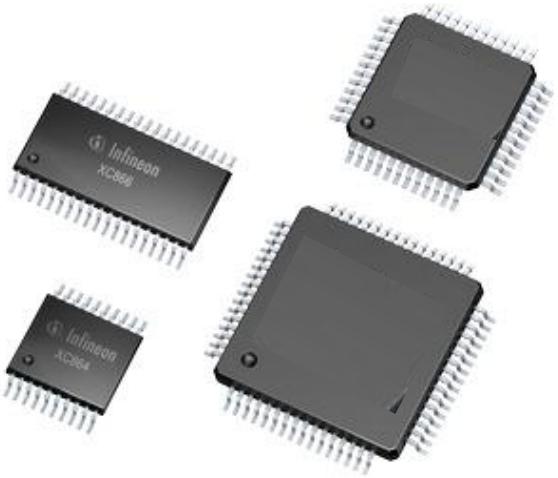
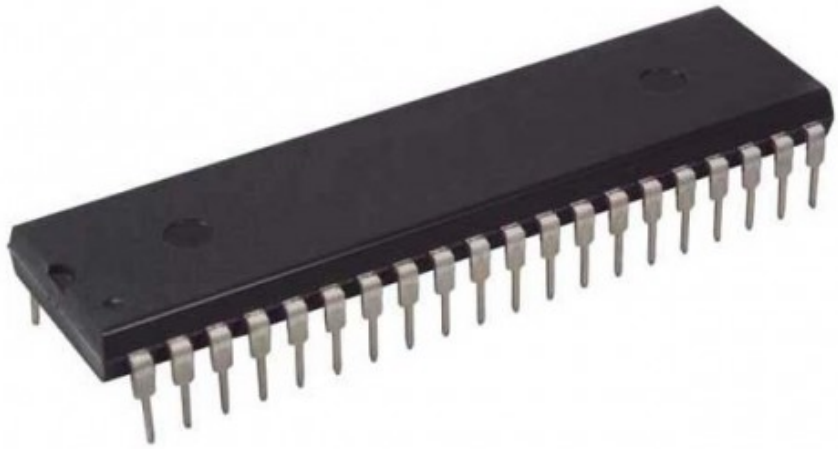
8051 Microcontroller

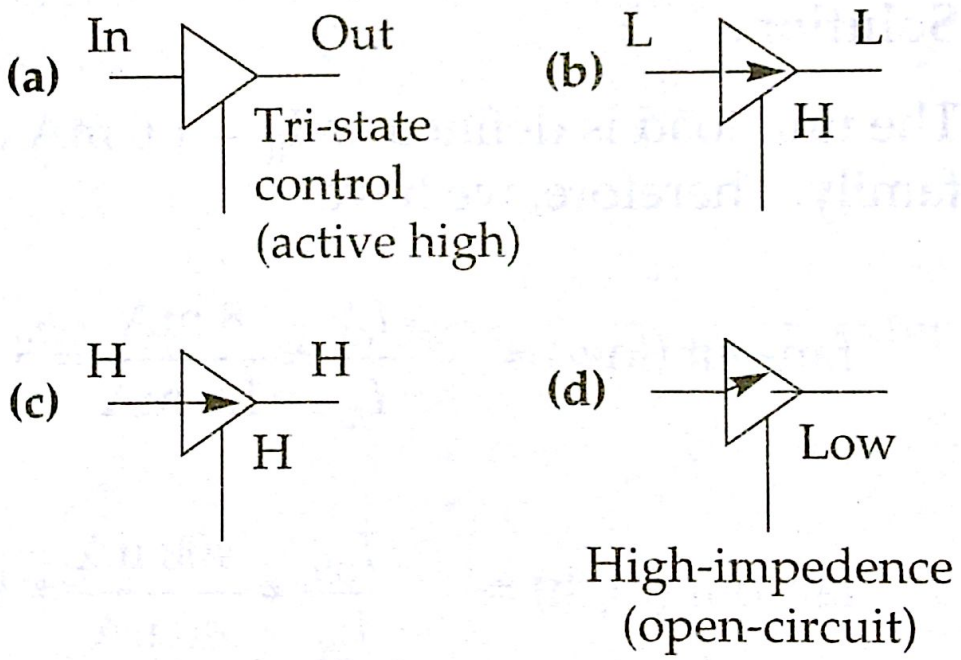




Internal structure of ports of 8051

- Since all ports of 8051 are bidirectional, they all have following 3 components in their structure
  1. D latch
  2. Output driver
  3. Input buffer

Tristate Buffer



- Tristate buffer has single input, single output and enable control pin.
- By activating the enable, data at input is transferred to output.
- When receiver current requirements exceeds the driver's capability, we must use buffer/drivers.
- Increase the fan out capacity of microcontroller
- When connecting IC chip to microcontroller, we need to find out how many input pins of IC chips can be driven by single output pin of microcontroller.

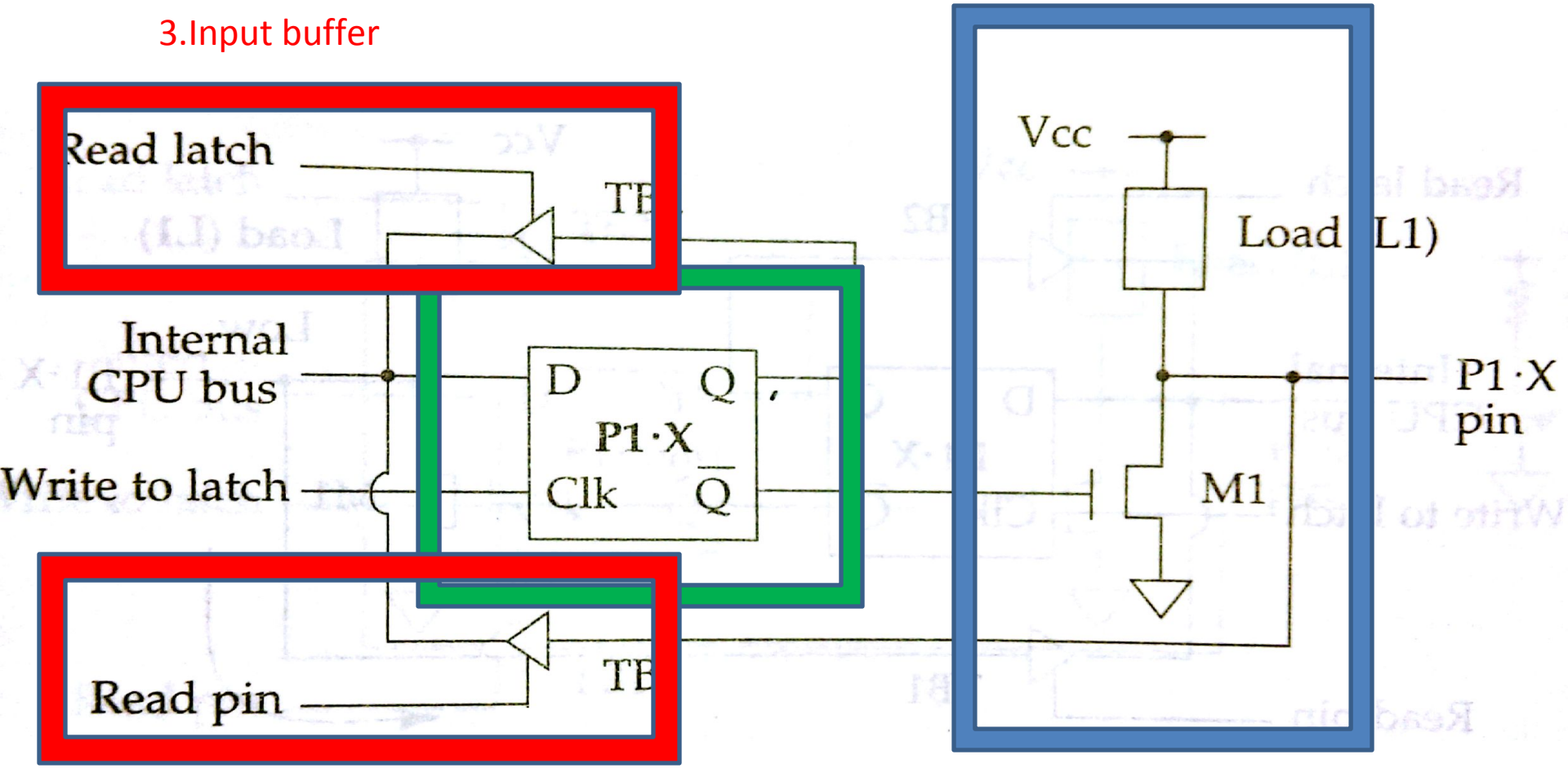
Internal structure of ports of 8051

- Since all ports of 8051 are bidirectional, they all have following 3 components in their structure

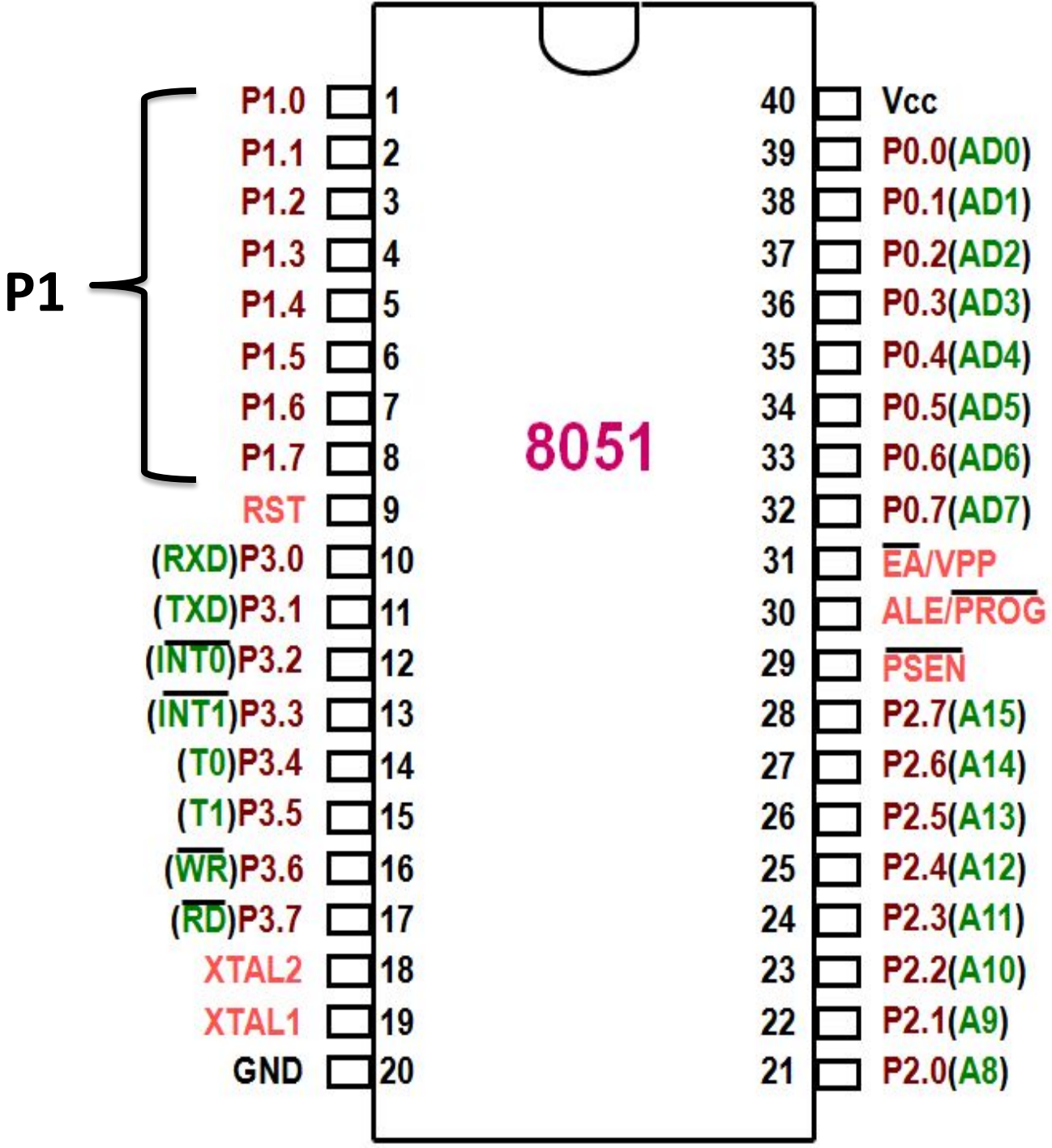
1.D latch

2.Output driver

3.Input buffer

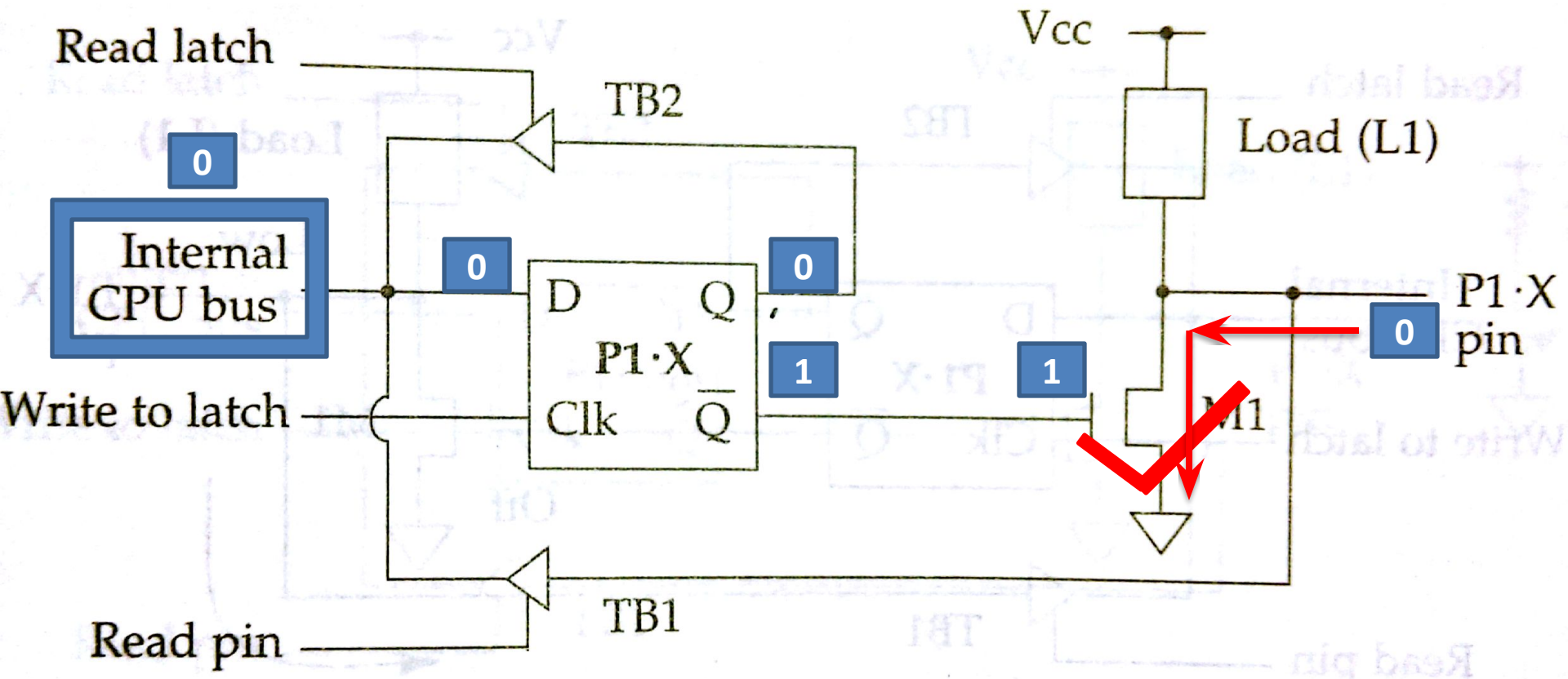


PORT 1

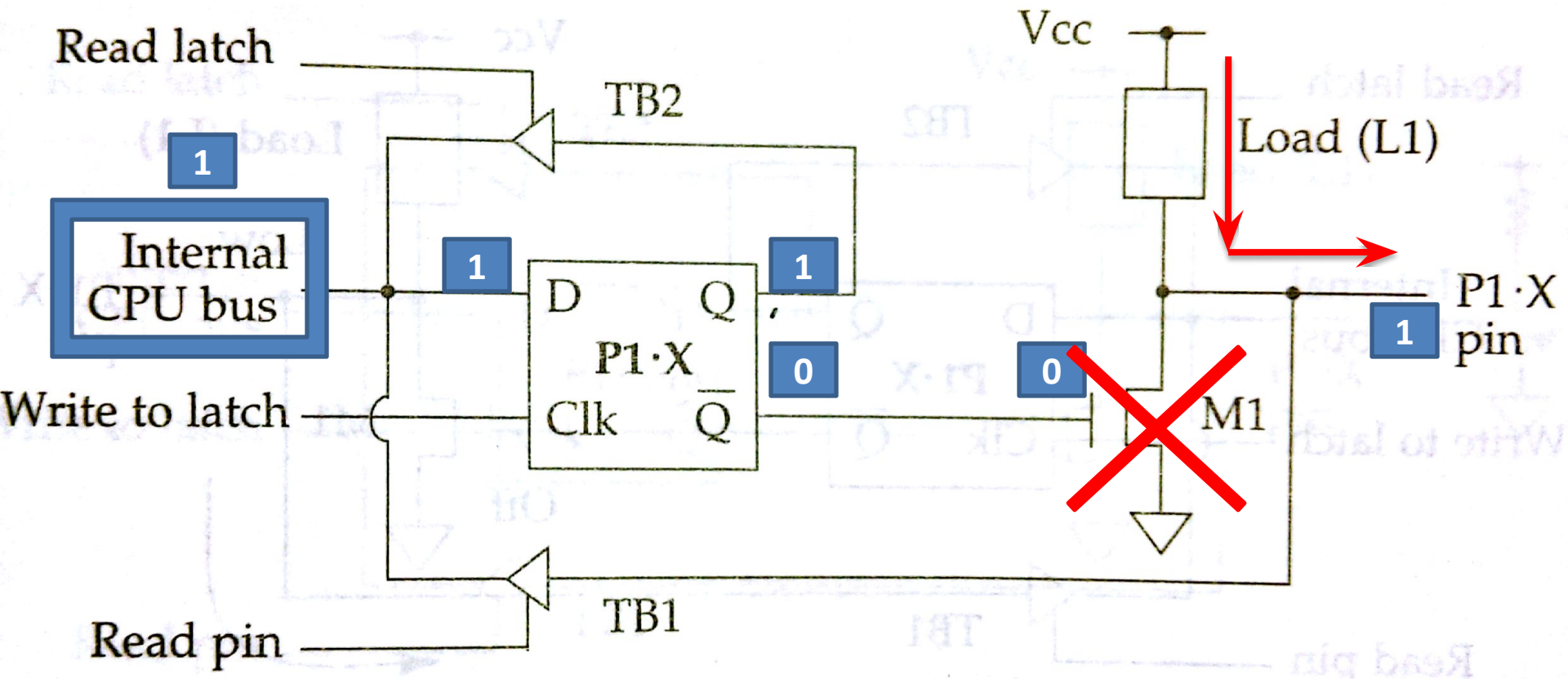

P1
8051
P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
RST 9
(RXD)P3.0 10
(TXD)P3.1 11
(INT0)P3.2 12
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
40 Vcc
39 P0.0(AD0)
38 P0.1(AD1)
37 P0.2(AD2)
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14)
26 P2.5(A13)
25 P2.4(A12)
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)

Port-1 does not have any alternate function i.e. it is dedicated solely for I/O interfacing. When used as output port, the pin is pulled up or down through internal pull-up. To use port-1 as input port, '1' has to be written to the latch. In this input mode when '1' is written to the pin by the external device then it read fine. But when '0' is written to the pin by the external device then the external source must sink current due to internal pull-up. If the external device is not able to sink the current the pin voltage may rise, leading to a possible wrong reading.

How to write data to port pin?



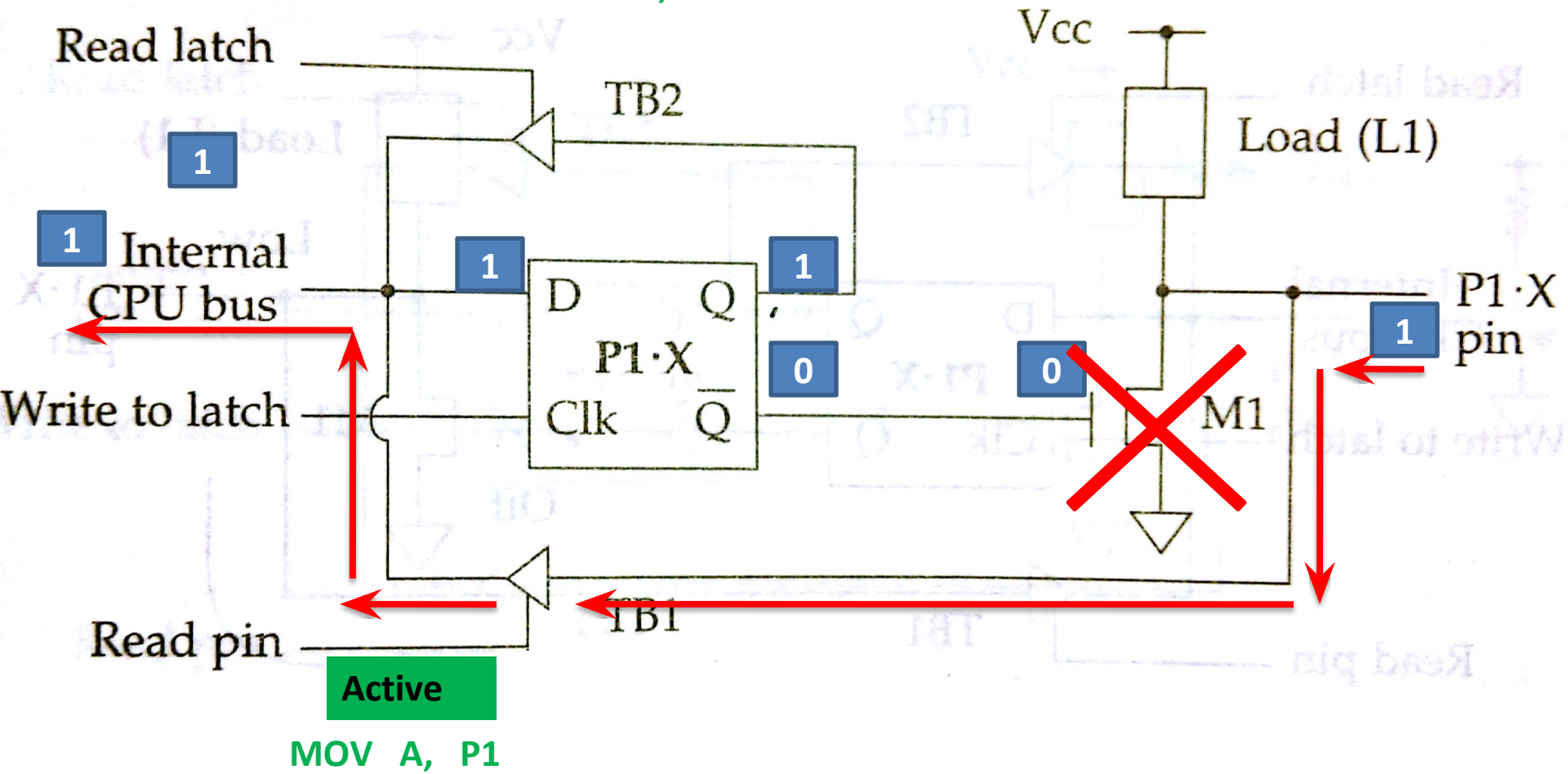
How to write data to port pin?



How to read port pin?

MOV P1, #0FFH; to configured port as input port, we must write 1(logic high) to that ports

MOV A, P1

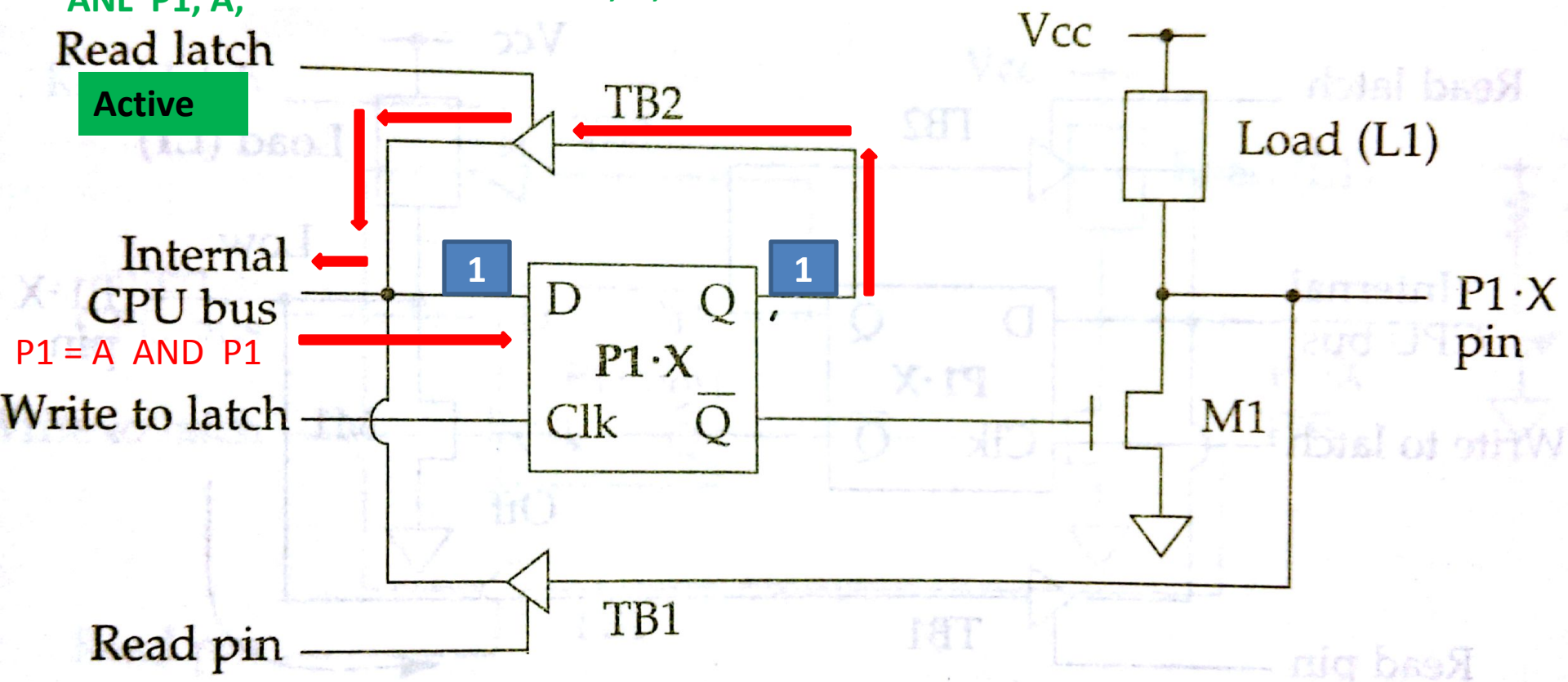


MOV A, P1

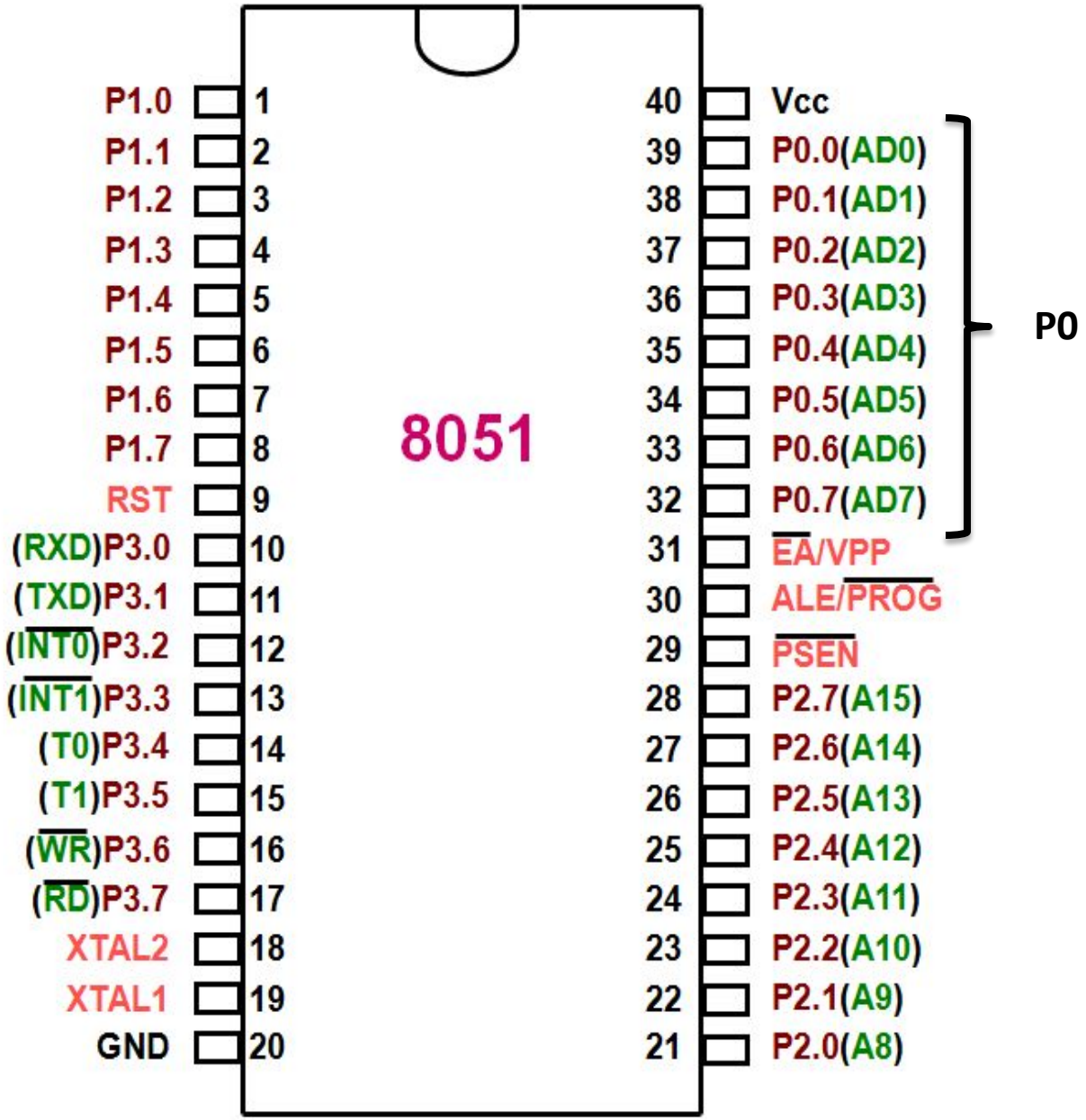
How to read latch?

MOV P1, #0FFH; to configured port as input port, we must write 1(logic high) to that ports

ANL P1, A; P1 = A AND P1



PORT 0


P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
RST 9
(RXD)P3.0 10
(TXD)P3.1 11
(INT0)P3.2 12
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
8051
40 Vcc
39 P0.0(AD0)
38 P0.1(AD1)
37 P0.2(AD2)
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14)
26 P2.5(A13)
25 P2.4(A12)
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)
P0

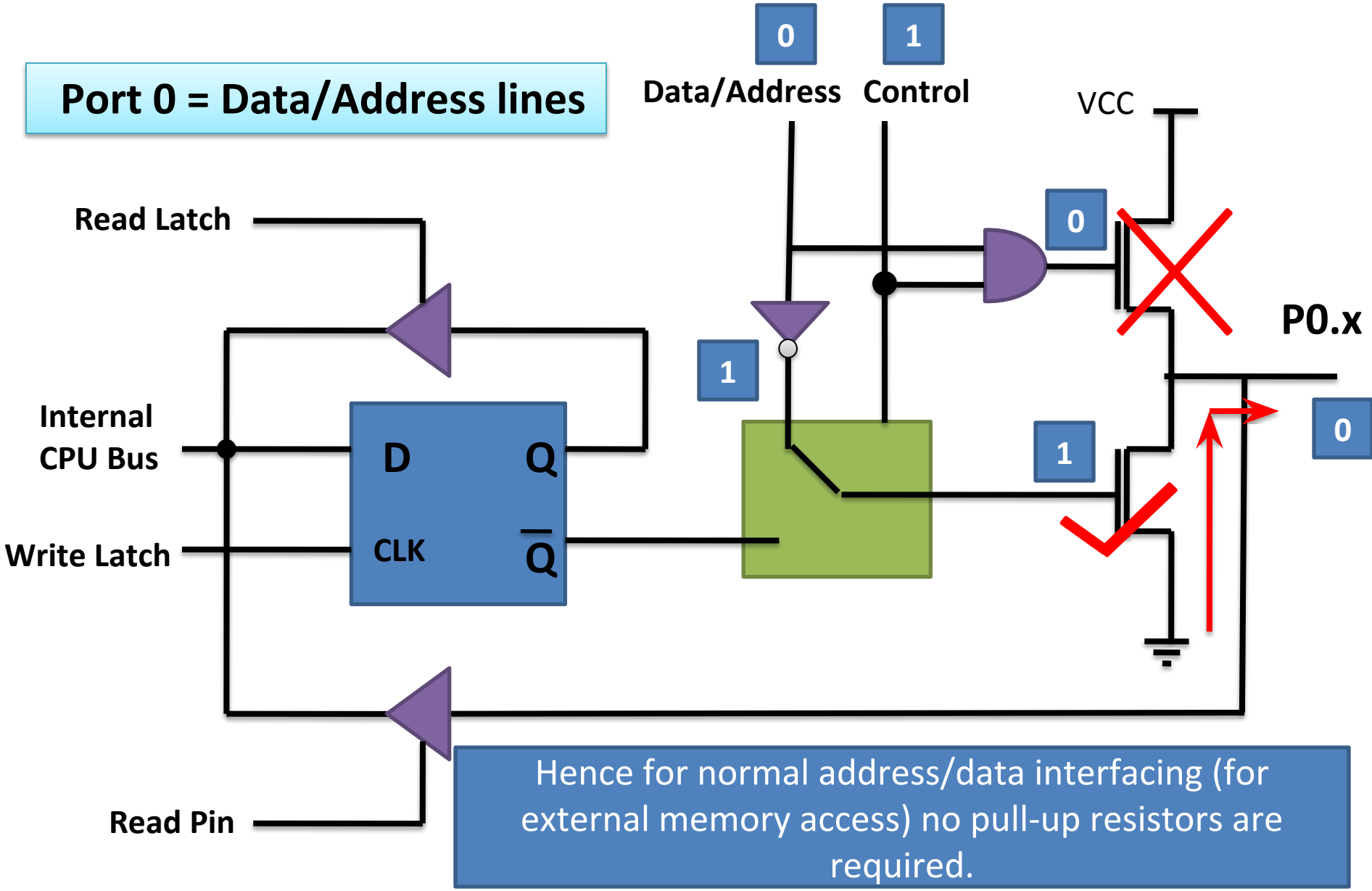
- Port-0 can be configured as a normal bidirectional I/O port or it can be used for address/data interfacing for accessing external memory.

- control =1 , the port is used for address/data interfacing.
- control =0, the port can be used as a normal bidirectional I/O port.

Let us assume that control is '0'. When the port is used as an input port, '1' is written to the latch. In this situation both the output MOSFETs are 'off'. Hence the output pin floats. This high impedance pin can be pulled up or low by an external source. When the port is used as an output port, a '1' written to the latch again turns 'off' both the output MOSFETs and causes the output pin to float. An external pull-up is required to output a '1'. But when '0' is written to the latch, the pin is pulled down by the lower MOSFET. Hence the output becomes zero.

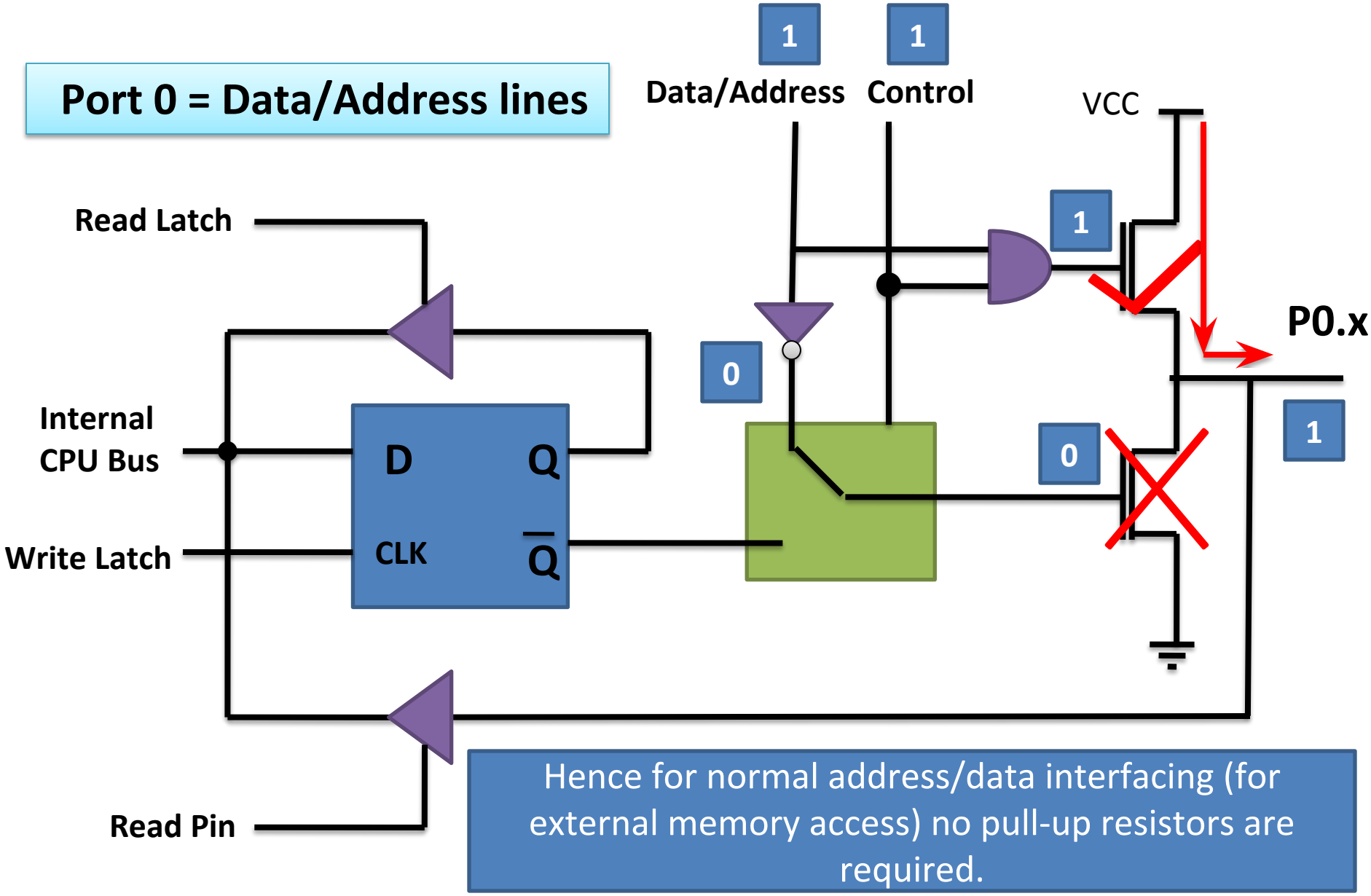
When the control is '1', address/data bus controls the output driver MOSFETs. If the address/data bus (internal) is '0', the upper MOSFET is 'off' and the lower MOSFET is 'on'. The output becomes '0'. If the address/data bus is '1', the upper transistor is 'on' and the lower transistor is 'off'. Hence the output is '1'. Hence for normal address/data interfacing (for external memory access) no pull-up resistors are required

Port 0 = Data/Address lines



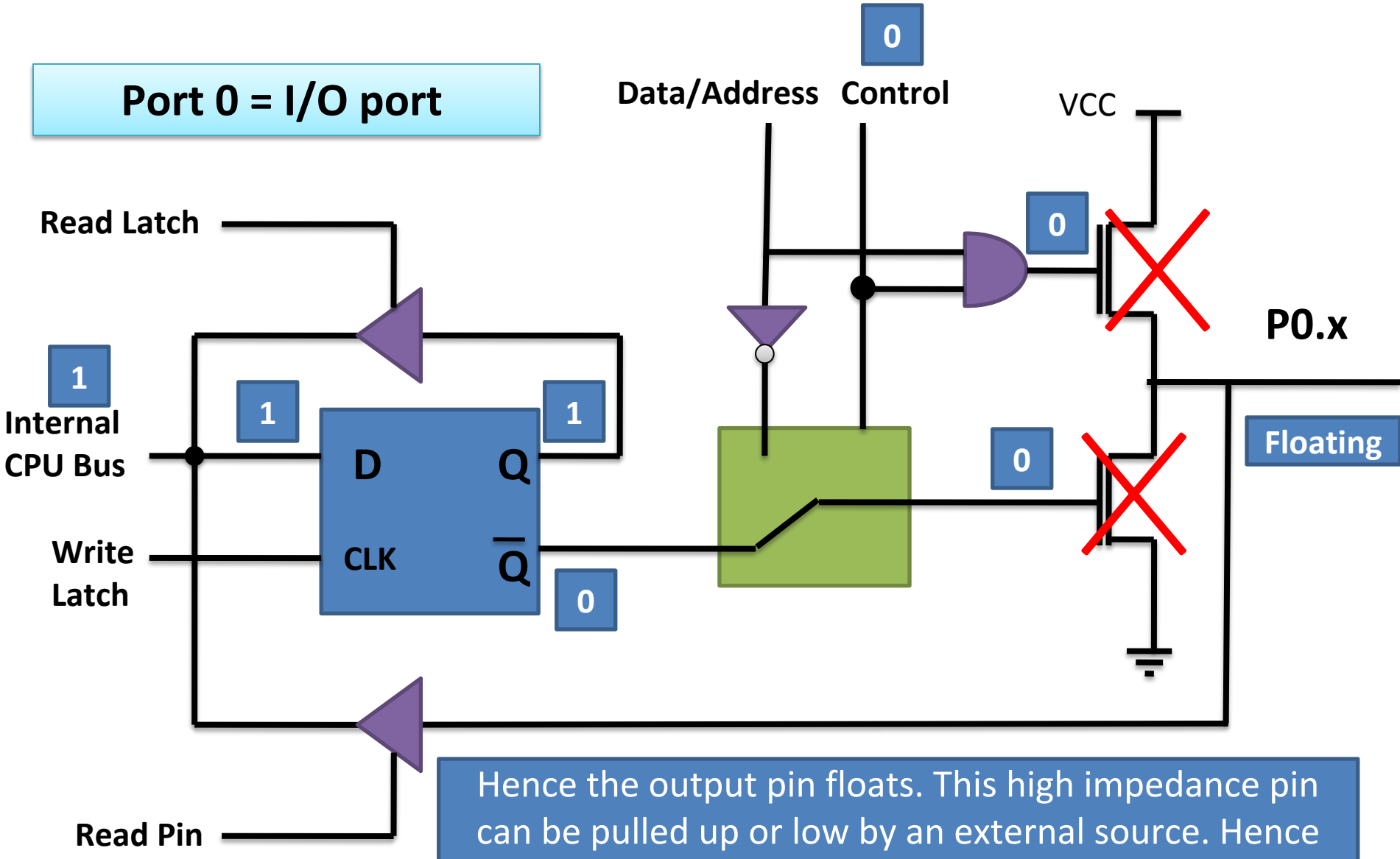
Hence for normal address/data interfacing (for external memory access) no pull-up resistors are required.

Port 0 = Data/Address lines



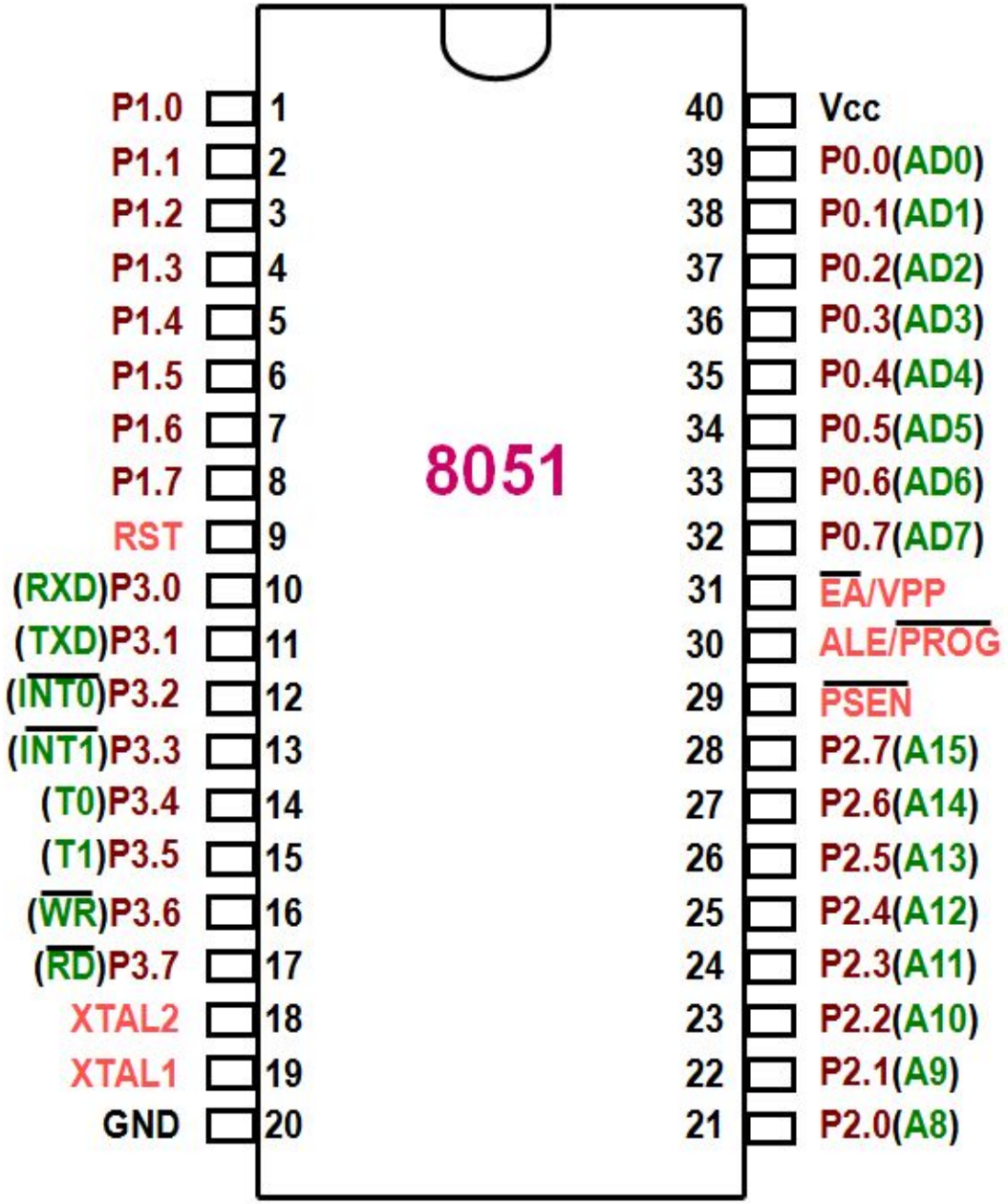
Hence for normal address/data interfacing (for external memory access) no pull-up resistors are required.

Port 0 = I/O port



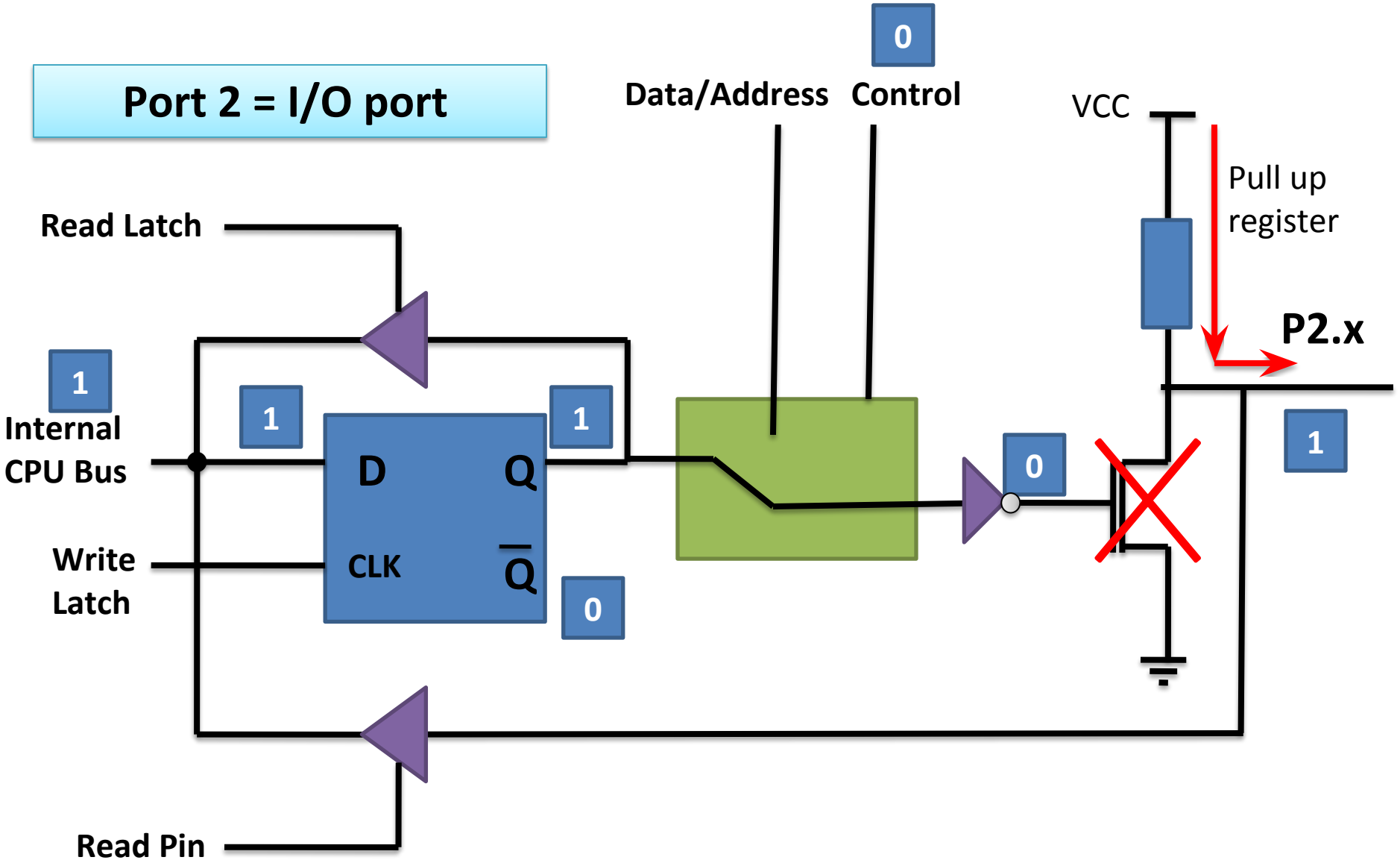
Hence the output pin floats. This high impedance pin can be pulled up or low by an external source. Hence pull up register required

PORT 2

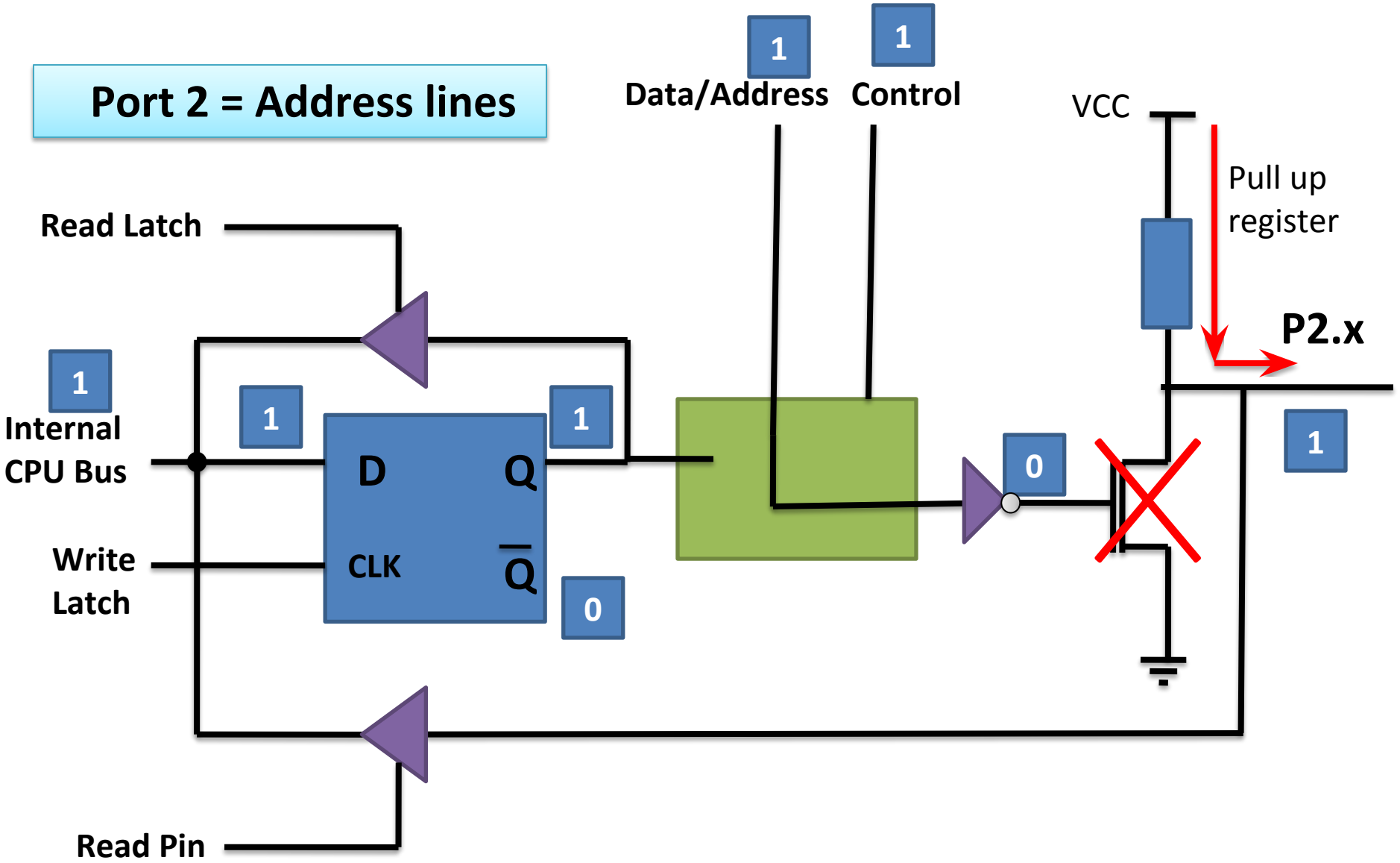

8051
P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
RST 9
(RXD)P3.0 10
(TXD)P3.1 11
(INT0)P3.2 12
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
40 Vcc
39 P0.0(AD0)
38 P0.1(AD1)
37 P0.2(AD2)
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14)
26 P2.5(A13)
25 P2.4(A12)
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)
P2

Port-2 is used for higher external address byte or a normal input/output port. The I/O operation is similar to Port-1. Port-2 latch remains stable when Port-2 pin are used for external memory access. Here again due to internal pull-up there is limited current driving capability.

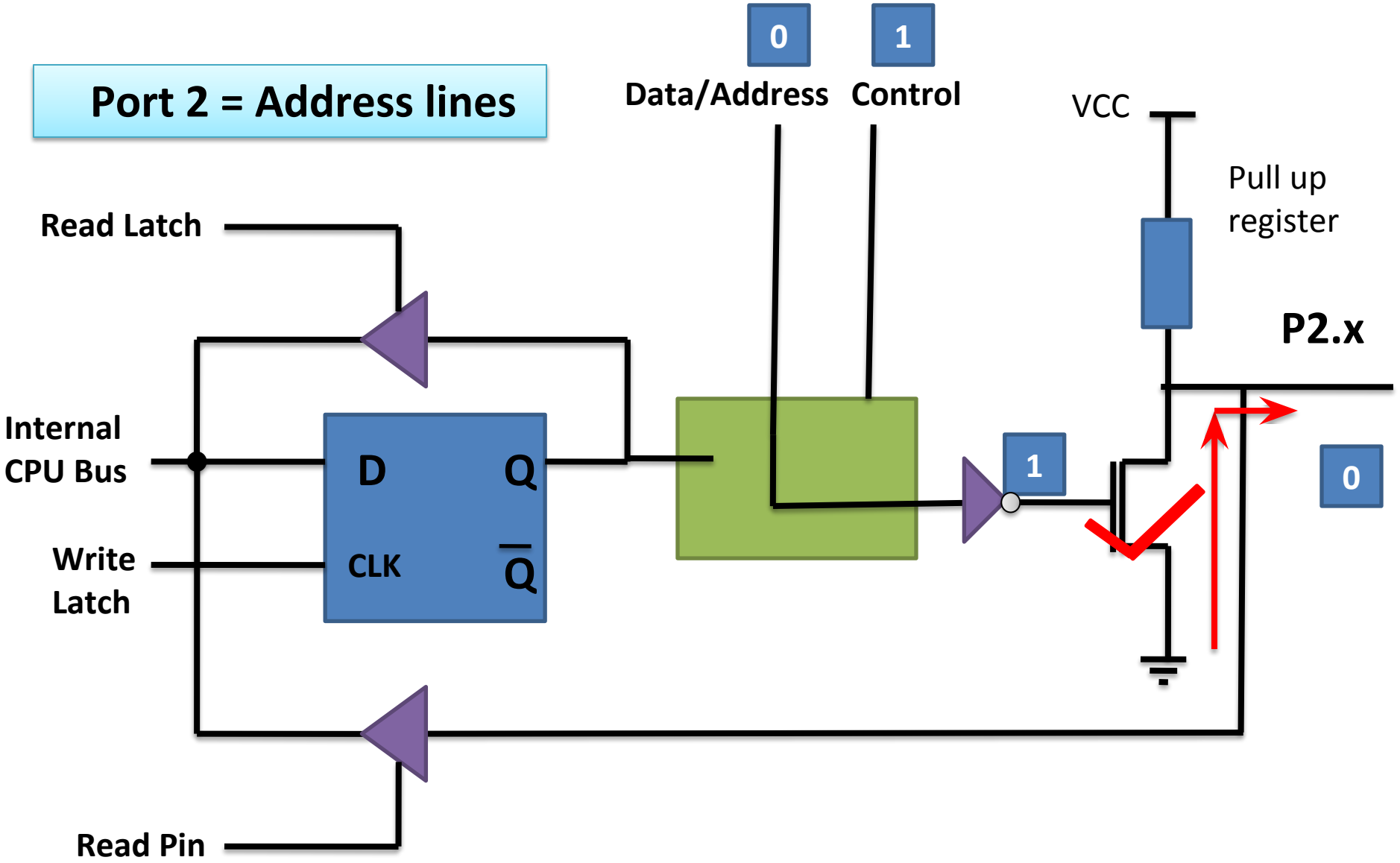
Port 2 = I/O port



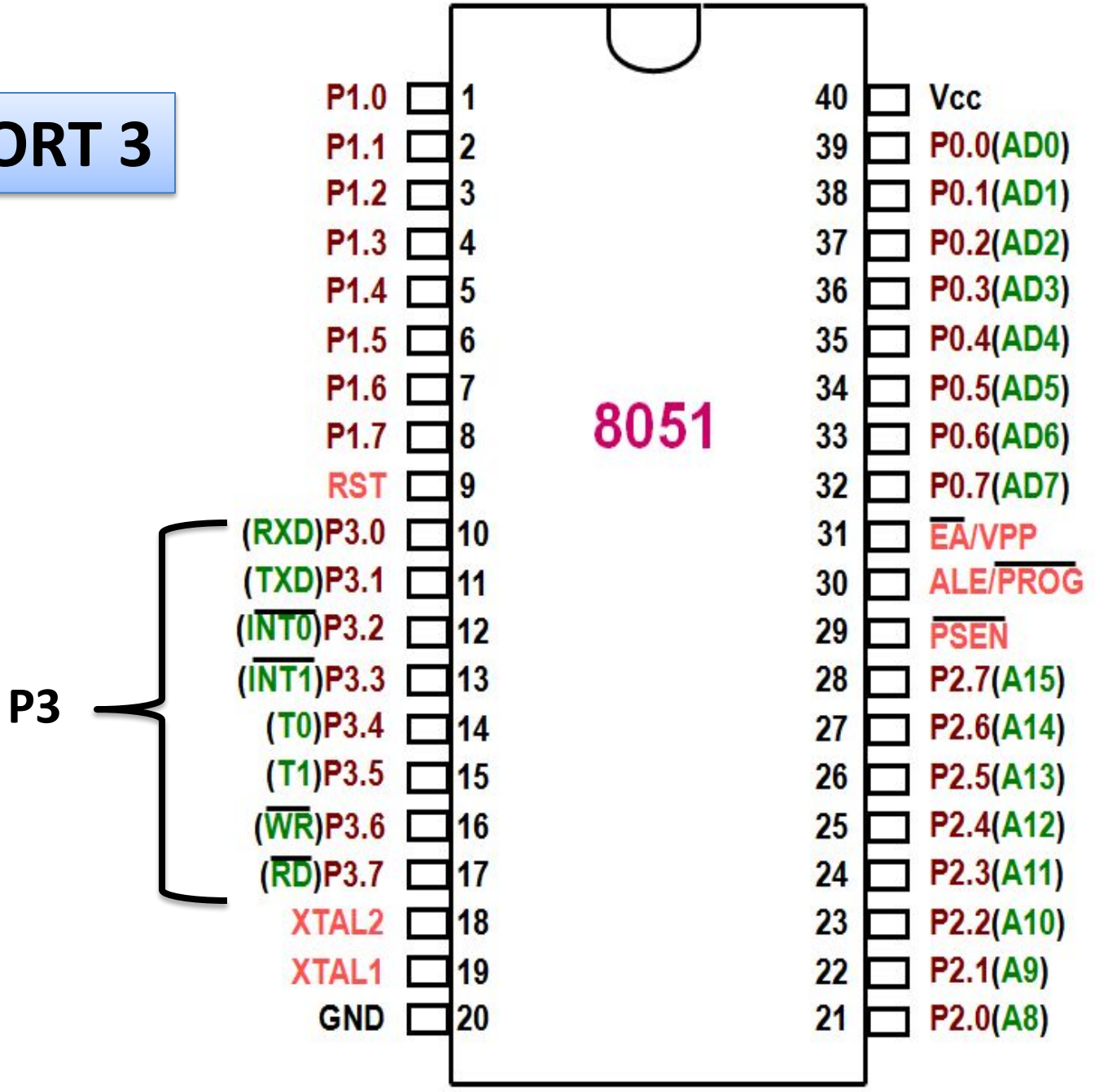
Port 2 = Address lines



Port 2 = Address lines



PORT 3


8051
P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
RST 9
(RXD)P3.0 10
(TXD)P3.1 11
(INT0)P3.2 12
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
40 Vcc
39 P0.0(AD0)
38 P0.1(AD1)
37 P0.2(AD2)
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14)
26 P2.5(A13)
25 P2.4(A12)
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)
P3

Each pin of Port-3 can be individually programmed for I/O operation or for alternate function. The alternate function can be activated only if the corresponding latch has been written to '1'. To use the port as input port, '1' should be written to the latch. This port also has internal pull-up and limited current driving capability.

Port 3 = I/O / Address lines

